

Thin film CPA for Boron doped silion

This set of notes continues the discussion on the paper [Ultrathin broadband nearly perfect absorber with symmetrical coherent illumination](https://opg.optica.org/viewmedia.cfm?seq=0&uri=oe-20-3-2246) (<https://opg.optica.org/viewmedia.cfm?seq=0&uri=oe-20-3-2246>).

We first start with the CPA condition, and the expressions for the reflection and transmission coefficients, $r(\omega)$, $t(\omega)$:

$$e^{inkd} = \pm \frac{n-1}{n+1} \text{ (Equation 4),}$$

$$r(\omega) = \frac{(n^2 - 1)(e^{2inkd} - 1)}{(n+1)^2 - (n-1)^2 e^{2inkd}} \text{ (Equation 2),}$$

$$t(\omega) = \frac{4ne^{inkd}}{(n+1)^2 - (n-1)^2 e^{2inkd}} \text{ (Equation 3)}$$

By introducing the expression for e^{inkd} (Equation 4) into the expressions for the reflection and transmission coefficients (Equations 2 and 3), we obtain the following:

$$\begin{aligned} r(\omega) &= \frac{(n^2 - 1)[(n-1)^2 - (n+1)^2]}{(n+1)^4 - (n-1)^4} = \\ &= -\frac{(n^2 - 1)[(n+1)^2 - (n-1)^2]}{[(n+1)^2 + (n-1)^2][(n+1)^2 - (n-1)^2]} = -\frac{(n^2 - 1)}{2(n^2 + 1)}, \end{aligned}$$

which is exactly **Equation 5**. As well as:

$$\begin{aligned} t(\omega) &= \frac{\pm 4n(n-1)(n+1)}{(n+1)^4 - (n-1)^4} = \pm \frac{4n(n^2 - 1)}{[(n+1)^2 - (n-1)^2][(n+1)^2 + (n-1)^2]} \\ &= \pm \frac{4n(n^2 - 1)}{[4n][2n^2 + 2]} = \pm \frac{(n^2 - 1)}{2(n^2 + 1)}, \end{aligned}$$

which is exactly **Equation 6**.

Thin film limit

We consider the regime $|nkd| \ll 1$ in which case the CPA condition reads:

$$1 + inkd = \pm \frac{n-1}{n+1} = \pm \left[1 - \frac{2}{n+1} \right]$$

The paper at this point approximates $n+1 \approx n$. I don't know how valid this is so we don't make this approximation (at least not yet). For the $+$ sign, we have (taking $n = n_1 + in_2$):

$$inkd = -\frac{2}{n+1} \rightarrow ikd(n_1 + 1 + in_2)(n_1 + in_2) = -2 \rightarrow$$

$$ikd(n_1 + in_2 + n_1^2 - n_2^2 + 2in_1n_2) = -2$$

The imaginary part of the LHS is proportional to $n_1 + n_1^2 - n_2^2$, which should be zero at CPA.

The real part of the equation, on the other hand, is given by

$$kd(-n_2 - 2n_1n_2) = -2 \rightarrow n_2 = 2/((2n_1 + 1)kd)$$

At this point, we see if there are solutions for $\{|n_1|, |n_2|\}kd \ll 1$ but, crucially, with $|n_1|, |n_2| \gg 1$, in which case we get $n_1 = n_2$:

$$n_1 = 1/(n_1kd) \rightarrow n_1 = n_2 = 1/\sqrt{kd}$$

Since in the thin film limit we have $d \ll \lambda = 2\pi/k \rightarrow kd \ll 1$ we have that $n_1 = n_2 \gg 1$ which means that n_1 is negligible compared to n_1^2 , which validates this approximation. Using $k = \omega/c$, we have **Equation 7**:

$$n_1 = n_2 = \frac{1}{\sqrt{kd}} = \sqrt{\frac{c}{\omega d}}$$

At what frequencies can we have $n_1 = n_2$? Since $n_1 + in_2 = \sqrt{\epsilon}$, if $\epsilon_1 = 0$, we'll have $\sqrt{\epsilon} = e^{i\pi/4} \sqrt{\epsilon_2} \rightarrow n_1 = n_2 = \sqrt{\epsilon_2/2}$. In the low frequency ($\omega \ll 1/\tau$) limit, we have:

$$\varepsilon = 1 - \frac{\omega_p^2}{(\omega^2 + 1/\tau^2)} + \frac{i\omega_p^2(1/\tau)}{\omega(\omega^2 + 1/\tau^2)} \approx 1 - \omega_p^2\tau^2 + i\omega_p^2\tau/\omega \approx i\omega_p^2\tau/\omega$$

$$\rightarrow n_1 \approx n_2 \approx \sqrt{\omega_p^2\tau/2\omega}$$

From this, we can obtain the thickness (**Woltersdorff thickness**) for CPA:

$$d = 2c/\omega_p^2\tau$$

We have another frequency at which $\varepsilon_1 \ll \varepsilon_2$: the plasmon frequency. At this frequency, we have $\varepsilon_\infty - \omega_p^2/(\omega^2 + 1/\tau^2) = 0 \rightarrow \omega^2 = \omega_p^2/\varepsilon_\infty - 1/\tau^2$. At this frequency, we have:

$$\varepsilon_2 = \frac{\varepsilon_\infty}{\sqrt{\tau^2\omega_p^2/\varepsilon_\infty - 1}} \rightarrow n_1 = n_2 = \sqrt{\frac{\varepsilon_\infty}{2\sqrt{\tau^2\omega_p^2/\varepsilon_\infty - 1}}}$$

At the plasmon frequency, then, we have a thickness for CPA given by:

$$d = 2c\tau/\varepsilon_\infty$$

Physical parameters for Boron doped silicon

We take a Drude model, with dielectric screening from bound electrons subsumed into $\varepsilon_\infty = 11.7$. The only additional ingredients required are the plasma frequency, ω_p and the carrier lifetime, τ . Both of these quantities may be determined from the carrier mobility, $\nu = e\tau/m$, where m is the effective mass. Let's find the units of ν in terms of volts. ν has units of $\text{charge} \times \text{distance}^2/\text{time}/\text{energy}$. Obviously, energy divided by charge has units of electric potential so we may write the mobility in units of $\text{cm}^2/(\text{second} \times \text{Volt})$

Once we have the mobility, we may obtain the Drude loss, $1/\tau = e/m\nu$. Furthermore, we may find the plasma frequency, $\omega_p^2 = N_c e^2/m\varepsilon_0$ using the relation,

$$\nu = \nu_{\min} + \frac{\nu_{\max} - \nu_{\min}}{1 + (N_c/N_r)^\alpha} \quad \text{(Equation 10)} \quad \text{(where we are provided all quantities except of } N_c), \text{ meaning:}$$

$$\left(\frac{N_c}{N_r}\right)^\alpha = \frac{\nu_{\max} - \nu}{\nu - \nu_{\min}} \rightarrow N_c = N_r \left[\frac{\nu_{\max} - \nu}{\nu - \nu_{\min}} \right]^{1/\alpha}$$

Estimate of plasma frequency and loss for heavily boron-doped silicon

We take $N_c = 4 \times 10^{19} \text{cm}^{-3}$. We write $e^2/\epsilon_0 = 4\pi\hbar c\alpha$ and $m = 0.26(0.511 \times 10^6 \text{eV}/c^2)$, which gives us:

$$\begin{aligned} \omega_p^2 &= 4 \times 10^{19} \text{cm}^{-3} \times 4\pi\hbar c^3 / (137 \times 0.26 \times 0.511 \times 10^6) = \\ 4 \times 10^{19} \times 4 \times \pi \times (20 \times 10^{-6}) \times (9 \times 10^{20}) / (137 \times 0.26 \times 0.511 \times 10^6) \\ &= (7.05 \times 10^{14})^2 \text{s}^{-1} \end{aligned}$$

To find the scattering time, we first calculate the mobility:

$$\begin{aligned} \nu &= 44.9 + \frac{470.5 - 44.9}{1 + (4 \times 10^{19})^{0.719} / (2.23 \times 10^{17})^{0.719}} \text{cm}^2 / (\text{V} \times \text{s}) = \\ &44.9 + \frac{425.6}{42.73} = 54.86 \text{cm}^2 / (\text{V} \times \text{s}) \end{aligned}$$

Therefore, we have:

$$\begin{aligned} \tau &= m\nu/e = (0.26 \times 0.511 \times 10^6 \times 1.60218 \times 10^{-19} \text{Joules}) / (9 \times 10^{16} \text{meter}^2 / \text{second}^2) \\ &\quad \times (54.86 \times 10^{-4} \text{meter}^2) / (1.602 \times 10^{-19} \text{Joules}) = \\ &\quad \frac{(0.26 \times 0.511 \times 10 \times 54.86)}{9} \text{femtoseconds} = 8.09 \text{femtoseconds} \end{aligned}$$

Woltersdorff thickness estimate

The Woltersdorff thickness is given by:

$$d = 2c/\omega_p^2 \tau = 6 \times 10^{17} \text{nm} / (8.09 \times 10^{-15} \times 7.05^2 \times 10^{28}) = 10^4 \times 6/8.09/7.05^2 = 149 \text{nanometers}$$

Plasmon thickness

The plasmon thickness is given by:

$$2c\tau/\epsilon_\infty = (6 \times 10^{17} \times 8.09 \times 10^{-15} / 11.7) \text{nanometers} = 414.87 \text{nanometers}$$

Conservation of energy

The Poynting vector is given by $\tilde{\mathbf{E}}(\mathbf{r}, t) \times \tilde{\mathbf{H}}(\mathbf{r}, t)$, where $\tilde{\mathbf{E}}(\mathbf{r}, t)$, $\tilde{\mathbf{H}}(\mathbf{r}, t)$ are the physical fields, given by the real parts of the fields derived above, i.e. they are given by:

$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \frac{1}{2} \left[\mathbf{E}(\mathbf{r}, \omega) e^{-i\omega t} + \mathbf{E}^*(\mathbf{r}, \omega) e^{i\omega t} \right] \text{ (and analogously for } \tilde{\mathbf{H}})$$

Therefore, the (time-averaged) Poynting vector may be written as:

$$\frac{1}{4} \left[\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega) + \mathbf{E}^*(\mathbf{r}, \omega) \times \mathbf{H}(\mathbf{r}, \omega) \right] = \frac{1}{2} \Re \left[\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega) \right]$$

Similarly, we have that the physical current in the thin film is given by:

$$\mathbf{J}(\mathbf{r}, t) = \frac{1}{2} \left[\sigma(\omega) \mathbf{E}(\mathbf{r}, \omega) e^{-i\omega t} + \sigma^*(\omega) \mathbf{E}^*(\mathbf{r}, \omega) e^{i\omega t} \right], \text{ so the time averaged dissipation}$$

in the material is given by:

$$\frac{1}{2} \int_{\Omega} \Re(\sigma(\omega)) |\mathbf{E}(\mathbf{r}, \omega)|^2 d\Omega$$

The power per unit area is given by $\frac{1}{2} \Re(\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega))$. We wish to verify that the net power flux into the thin film is equal to the power dissipated by the moving charges in the thin film. In the following, we consider all quantities per unit area since we have translational symmetry in the plane of the thin film. We consider the most general case, with incoming plane waves of amplitude A and B , respectively. The field to the left of the thin film is given by:

$$\mathbf{E}(\mathbf{r}, \omega) = \left[A e^{ik(x+d/2)} + (rA + tB) e^{-ik(x+d/2)} \right] \mathbf{e}_y$$

$$\mathbf{H}^*(\mathbf{r}, \omega) = \frac{i}{\omega \mu_0} \partial_x (\mathbf{E}^*(\mathbf{r}, \omega) \cdot \mathbf{e}_y) \mathbf{e}_z =$$

$$\left[\frac{k}{\omega \mu_0} \left(A^* e^{-ik(x+d/2)} - (r^* A^* + t^* B^*) e^{ik(x+d/2)} \right) \right] \mathbf{e}_z$$

Therefore, we have:

$$\frac{1}{2} \Re(\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega)) = \frac{k}{2\omega\mu_0} (|A|^2 - |rA + tB|^2) \mathbf{e}_x$$

Analogously, we may find the power entering the material from the right side of the thin film: $(k/2\omega\mu_0)(|B|^2 - |tA + rB|^2)$, giving us a total net flux into the thin film given by:

$$\frac{1}{2c\mu_0} (|A|^2 + |B|^2 - |rA + tB|^2 - |tA + rB|^2),$$

where we also used $k = \omega/c$. The power dissipated in the material (taking E_1, E_2 to be the amplitudes of the right and leftgoing waves within the material), per unit volume, is given by:

$$\begin{aligned} & \frac{1}{2} \Re(\sigma(\omega)) [E_1 e^{ipx} + E_2 e^{-ipx}] [E_1^* e^{-ip^*x} + E_2^* e^{ip^*x}] = \\ & \frac{1}{2} \Re(\sigma(\omega)) [|E_1|^2 e^{-2p_2x} + |E_2|^2 e^{2p_2x} + E_1 E_2^* e^{2ip_1x} + E_1^* E_2 e^{-2ip_1x}], \end{aligned}$$

where we have defined $p = p_1 + ip_2$. This gives for the power dissipated per unit area:

$$\frac{1}{2} \Re\sigma(\omega) \left[(|E_1|^2 + |E_2|^2) \left[\frac{e^{p_2d} - e^{-p_2d}}{2p_2} \right] + (E_1 E_2^* + E_1^* E_2) \left[\frac{e^{ip_1d} - e^{-ip_1d}}{2ip_1} \right] \right]$$

We know that $\varepsilon = 1 + i\sigma/\omega\varepsilon_0 \rightarrow \Re\sigma = \omega\varepsilon_0\varepsilon'' = 2\omega\varepsilon_0 n_1 n_2 = 2\omega/(c^2\mu_0)$

Therefore, we may rewrite the above as:

$$\frac{1}{2c\mu_0} \left[n_1 (|E_1|^2 + |E_2|^2) [e^{p_2d} - e^{-p_2d}] - in_2 (E_1 E_2^* + E_1^* E_2) [e^{ip_1d} - e^{-ip_1d}] \right]$$

Therefore, our goal is to verify the following equality:

$$\begin{aligned} & \left[n_1 (|E_1|^2 + |E_2|^2) [e^{p_2d} - e^{-p_2d}] - in_2 (E_1 E_2^* + E_1^* E_2) [e^{ip_1d} - e^{-ip_1d}] \right] = \\ & \quad (|A|^2 + |B|^2 - |rA + tB|^2 - |tA + rB|^2) \end{aligned}$$

E_1, E_2 are implicitly given by:

$$A(1+r) + tB = E_1 e^{-ipd/2} + E_2 e^{ipd/2}$$

$$B(1+r) + tA = E_1 e^{ipd/2} + E_2 e^{-ipd/2}$$

as well as by:

$$(1-r)A - tB = n(E_1 e^{-ipd/2} - E_2 e^{ipd/2})$$

$$tA - (1-r)B = n(E_1 e^{ipd/2} - E_2 e^{-ipd/2})$$

Let's multiply the second equation by the complex conjugate of the fourth equation. The LHS will be given by:

$$|tA|^2 - |B|^2(1+r-r^*-|r|^2) + A^* B t^*(1+r) - B^* A(1-r^*)t$$

Let's multiply the first equation by the complex conjugate of the third equation. The LHS will be given by:

$$-|tB|^2 + |A|^2(1-r^*+r-|r|^2) + t(1-r^*)BA^* - (1+r)t^*AB^*$$

Let's subtract this from the first expression, giving us:

$$(|A|^2 + |B|^2)(|t|^2 + |r|^2 - r + r^* - 1) + (A^*B + B^*A)(t^* - t + t^*r + r^*t)$$

Now, note that the negation of the real part of this expression is the RHS of the equality we want to demonstrate. Therefore, we repeat all the same steps for the RHS's of the boundary condition equations. Multiplying the second BC equation by the complex conjugate of the fourth BC equation gives us:

$$n^*(E_1 e^{ipd/2} + E_2 e^{-ipd/2})(E_1^* e^{-ip^*d/2} - E_2^* e^{ip^*d/2}) =$$

$$n^* \left[|E_1|^2 e^{-p_2d} - |E_2|^2 e^{p_2d} + E_1^* E_2 e^{-ip_1d} - E_1 E_2^* e^{ip_1d} \right]$$

Now, as before, let's multiply the first BC equation by the complex conjugate of the third BC equation. The right hand side is given by:

$$n^* \begin{bmatrix} E_1 e^{-ipd/2} + E_2 e^{ipd/2} \end{bmatrix} \begin{bmatrix} E_1^* e^{ip^*d/2} - E_2^* e^{-ip^*d/2} \end{bmatrix} =$$

$$n^* \begin{bmatrix} |E_1|^2 e^{p_2d} - |E_2|^2 e^{-p_2d} + E_1^* E_2 e^{ip_1d} - E_1 E_2^* e^{-ip_1d} \end{bmatrix}$$

We subtract this from the first expression, giving us:

$$n^* \begin{bmatrix} \left(|E_1|^2 + |E_2|^2 \right) \left(e^{-p_2d} - e^{p_2d} \right) + \left(E_1^* E_2 + E_2^* E_1 \right) \left(e^{-ip_1d} - e^{ip_1d} \right) \end{bmatrix}$$

We take the real part of this, which gives us:

$$n_1 \begin{bmatrix} |E_1|^2 + |E_2|^2 \end{bmatrix} \left(e^{-p_2d} - e^{p_2d} \right) - in_2 \begin{bmatrix} E_1^* E_2 + E_2^* E_1 \end{bmatrix} \left(e^{-ip_1d} - e^{ip_1d} \right),$$

which is the negation of the LHS of the equality which we wanted to prove—QED.